Xijia "Polina" Zhang

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Education

Georgia Institute of Technology Aug. 2024 – Dec. 2028
Ph.D. in Robotics (under the School of Computer Science)

University of Michigan Sep. 2022 – May. 2024
B.S.E in Computer Science

Shanghai Jiao Tong University Sep. 2020 – Aug. 2024
B.E in Electrical and Computer Engineering

Publications

"FlowEval: How Much Adaptation Does an Embodied Foundation Model Need at Deployment?"
Zhang Xi-Jia, Roman Yakunin, Shivang Chopra, Chengyue Huang, Che Chen, Siddharth Karamcheti, Zsolt Kira *(in submission)*

– Propose a training free method to quantify the gap between pretrained embodied foundation models and downstream deployments, and predict the amount of adaptation data required.

"CAPEX: Efficiently Distilling Foundation Model Behavior into Deployable Robot Policies through Experience-Adaptive Reasoning"
Shivam Aarya, **Zhang Xi-Jia**, Chengyue Huang, Junhyun Kim, Huishu Xue, Hrishit Leen, Roman Yakunin, Animesh Garg, Zsolt Kira *(in submission)*

– Distill GPT-6-Astra's end-to-end joint control into lightweight policies per task.
– Develop an agentic framework that generates actions in chunks and learns from physical interaction experience.

"Model-Agnostic Policy Explanations with Large Language Models"
Zhang Xi-Jia, Yue Guo, Shufei Chen, Simon Stepputtis, Matthew Gombolay, Katia Sycara, Joseph Campbell COLM 2025 (**top 5%**)

– Propose a framework for generating language explanations of agent behavior based on observed states and actions. Build and ablate different harnesses.

"Communication and Verification in LLM Agents towards Collaboration under Information Asymmetry
Run Peng, Ziqiao Ma, Amy Pang, Sikai Li, **Zhang Xi-Jia**, Yingzhuo Yu, Cristian-Paul Bara, Joyce Chai COLM 2026

– Investigate the performance of LLM agents in human-robot collaboration tasks under information asymmetry.

"Learning Effective Action Advising in the Face of Changing Rewards"
Yue Guo, **Zhang Xi-Jia**, Simon Stepputtis, Joseph Campbell, Katia Sycara CoLLAs 2024 (**Oral**)

– Enable the teacher policy to continually learn and adapt its reward function through ongoing observation of the student when providing action advices.

"Sensor Array Optimization for the Electronic Nose via Different Deep Learning Methods"
Zhang Xi-Jia*, Tao Wang*, Wangze Ni, Yongwei Zhang, Wen Lv, Min Zeng, Jianhua Yang, Nantao Hu, Rui Zhan, Guang Li, Zhiqiang Hong, Zhi Yang Sensors and Actuators: B, 2024

– Compare lightweight machine learning models on an Electronic Nose and investigate how their performance scales with dataset size.

"Teaching the Teacher: Enhancing Human-to-Robot Skill Demonstration with Live Foundation Model and Augmented Reality Feedback"

Nina Moorman, Matthew Luebbers*, **Zhang Xi-Jia***, Zulfiqar Zaidi, Marcus Lau, Yixing Yao, Megan Langwasser, Letian Chen, Sanne van Waveren, Matthew Gombolay HRI 2026

– Support non-expert users in providing kinesthetic demonstrations through language explanations and policy visualizations.

Skills

`Real Robot Deployment`	TIAGo, Bimanual-YAM, Franka, Jaco Arm, Khepera, Fetch, OpenArm
`Simulation`	RViz, Gazebo, Mujoco
`Foundation Model`	Fine-tuning, Deploying, Harness; LoRA, PPO
`Toolchain`	ROS, ROS2, Linux, Docker, Huggingface, RLlib
`Frameworks & Libraries`	Gym, NLTK, Transformers, Pytorch, Scikit-Learn

Community Service

Reviewer ICLR 24-25, COLM 25-26, HRI 24-26

Honors & Awards

Sep. 2024	Robotics Fellowship	*Georgia Institute of Technology*
Mar. 2024	James B. Angell Scholar	*University of Michigan*
Apr. 2024	Elected Member	*Tau Beta Pi, Michigan Gamma*
2022-2023	Dean's List	*University of Michigan*
May. 2023	Chun-Tsung Scholar 150/13,000	*Shanghai Jiao Tong University*
Oct. 2021	Rongchang Scholarship Nomination 6/786	*Shanghai Jiao Tong University*
Nov. 2021	Silver Medal	*The University Physics Contest*